Stroke

CLINICAL AND POPULATION SCIENCES

Sex Differences in Stroke Care and Outcome 2005–2018

Observations From the Swedish Stroke Register

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BACKGROUND AND PURPOSE: Previous studies of stroke management and outcome in Sweden have revealed differences between men and women. We aimed to analyze if differences in stroke incidence, care, and outcome have altered over time.

METHODS: All stroke events registered in the Swedish Stroke Register 2005 to 2018 were included. Background variables and treatment were collected during the acute hospital stay. Survival data were obtained from the national cause of death register by individual linkage. We used unadjusted proportions and estimated age-adjusted marginal means, using a generalized linear model, to present outcome.

RESULTS: We identified 335 183 stroke events and a decreasing incidence in men and women 2005 to 2018. Men were on average younger than women (73.3 versus 78.1 years) at stroke onset. The age-adjusted proportion of reperfusion therapy 2005 to 2018 increased more rapidly in women than in men (2.3%–15.1% in men versus 1.4%–16.9% in women), but in 2018, women still had a lower probability of receiving thrombolysis within 30 minutes. Among patients with atrial fibrillation, oral anticoagulants at discharge increased more rapidly in women (31.2%–78.6% in men versus 26.7%–81.9% in women). Statins remained higher in men (36.9%–83.7% in men versus 32.3%–81.2% in women). Men had better functional outcome and survival after stroke. After adjustment for women's higher age, more severe strokes, and background characteristics, the absolute difference in functional outcome was <1% and survival did not differ.

CONCLUSIONS: Stroke incidence, care, and outcome show continuous improvements in Sweden, and previously reported differences between men and women become less evident. More severe strokes and older age in women at stroke onset are explanations to persisting differences.

GRAPHIC ABSTRACT: An online graphic abstract is available for this article.

Key Words: healthcare disparities ■ mortality ■ patient reported outcome ■ secondary prevention ■ sex ■ stroke ■ thrombolysis

In Sweden, the number of stroke patients has declined over the last decade, from around 30 000 persons per year to 25 000 persons yearly.[1] This notwithstanding an increase in population during the past 2 decades, from 9.0 million inhabitants in 2005 to 10.2 million in 2018,[2] and an increase in life expectancy, both in men (77.9–81.7 years) and in women (82.3–85.0 years).[3] Why the incidence of stroke is declining is probably multifactorial. Suggested explanations are better primary prevention and changes in risk factors in the population.[4,5] As a group, men have their stroke 5 years earlier than women; in the younger age groups men dominate, while women make up the majority in patients above 80 years.[1,6–8] Several studies report women to have more severe symptoms on arrival to hospital, worse prognosis, and lower quality of life after stroke compared with men[6,9–12] but other demonstrate contradictory results,[13] including a more favorable outcome.[14] The conflicting results may, at

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Nonstandard Abbreviations and Acronyms

ADL	activities in daily living
AF	atrial fibrillation
CF	case fatality
Riksstroke	The Swedish Stroke Register

least in part, be explained by differences, in study design, geographic area studied, time period, and focus of study.

In addition to access to stroke units, several studies have also shown differences between men and women in access to thrombolysis. This has been reported to be a consequence of a longer prehospital delay due to women being older and more often living alone at stroke onset.[7,15–17] Furthermore, although secondary preventive drug treatment is well known to lower the risk of recurrent stroke, differences between men and women in use of secondary preventive drugs have been described, particularly regarding anticoagulants and statins after ischemic stroke.[18–21]

The aim of this register-based, nation-wide study was to explore presence of sex-specific differences in stroke incidence, risk factors, acute treatment, secondary prevention, and outcome, and if present, analyze if these differences changed between 2005 and 2018.

METHODS

Material

Because of the sensitive nature of the data collected for this study, requests to access the data set from qualified researchers trained in human subject confidentiality protocols may be sent to the Swedish Stroke Register (Riksstroke) at riksstroke@regionvasterbotten.se.

This study included all patients, 18 years and older, treated in hospital and registered in Riksstroke with an intracerebral hemorrhage, ischemic stroke or an unspecified stroke (*International Classification of Diseases, Tenth Revision*: I61, I63, and I64) during 2005 to 2018. The study population was retrieved by Riksstroke.

Riksstroke is a national quality register, monitoring stroke care. All hospitals admitting acute stroke patients (72 hospitals during the study period) participate. Riksstroke is estimated to include over 90% of all acute stroke patients treated in hospital.[1] The register includes information on age, sex, living situation, and activities of daily living (ADL) before stroke, cardiovascular risk factors (diabetes, atrial fibrillation [AF], hypertensive treatment, and smoking), stroke diagnosis, stroke severity (fully conscious, drowsy, and unconscious), acute treatment, and secondary preventive drugs. Patients are followed-up by a questionnaire 3 months after stroke, in which patient-reported outcomes and experiences of care are registered. Information is entered into the registry at each hospital, with a few exceptions by a nurse. Riksstroke includes automatic data checks upon data entry, and the validity is continuously evaluated. The registry is opt-out. The response rate to the questionnaires is high, 83%.[1]

The incidence per 100 000 inhabitants was estimated based on the number of registrations in Riksstroke and the sex-specific size of population as of December 31 each year (from Statistics Sweden).

As in Riksstroke, we defined stroke unit care to include neurosurgical and intensive care unit. Reperfusion therapy, that is, thrombolysis and thrombectomy, was analyzed in patients with ischemic stroke only. As of 2018, Swedish national guidelines recommend thrombolysis, when indicated, to be started at least within 30 minutes after hospital arrival[22]; information on timing of thrombolysis was available in Riksstroke from 2010.

We studied secondary preventive drugs in patients discharged from hospital alive. Blood pressure-lowering drugs was analyzed in all patients and lipid-lowering drugs (ie, statins) in patients with ischemic stroke. Anticoagulants, including warfarin and nonvitamin K antagonist oral anticoagulants, were analyzed in patients with ischemic stroke and AF and antiplatelets in patients with ischemic stroke without AF.

Independency in ADL was defined by the ability to move around indoors, to manage dressing, and to use bathroom without assistance. We analyzed ADL dependency at 3 months after stroke in patients who were independent before their stroke.

Date of death was retrieved by individual linkage with the cause of death register (managed by the National Board of Health and Welfare) and used to calculate survival time and case fatality (CF) at 7, 30, and 90 days after stroke onset. The cause of death register includes all deaths occurring in Sweden.

Results were reported according to the Reporting of Studies Conducted Using Observational Routinely Collected Health Data Statement. A completed Reporting of Studies Conducted Using Observational Routinely Collected Health Data checklist and a flow diagram can be found in the Data Supplement (Table I and Figure I in the Data Supplement).

Because of the sensitive nature of the data collected for this study, requests to access the data set from qualified researchers trained in human subject confidentiality protocols may be sent to Riksstroke at riksstroke@regionvasterbotten.se.

Statistical Methods

The initial analyses of incidence, patient characteristics, cardiovascular risk factors, acute treatment, secondary prevention, and outcome were based on unadjusted proportions.

The stroke incidence was analyzed by a generalized linear model, using Poisson distribution, a log link function, and an offset equal to the natural logarithm of the size of the general population. The model included the independent variables year, sex, and the sex-by-year interaction. Another generalized linear model (logistic regression) was used to estimate age-adjusted proportions (marginal means) with 95% CIs. This model was also used to test if the overall proportions or the time trend differed between men and women. The model included the independent variables sex, year of stroke, age, age^2, and sex-by-year interaction. By including age2, the model allowed a nonlinear (U shaped) relationship between age and the response variable. The sex-by-year interaction in the above models tested if the relative difference between men and women had been constant during the study period (ie, if the time trend differed).

Table 1. Patient Characteristics, Acute Treatment, Secondary Prevention, and Outcome

			Sex			
	Overall		Men		Women	
	N	(%)	n	(%)	n	(%)
Patient characteristics						
Age, y						
18–54	20 280	(6.1)	12 749	(7.4)	7531	(4.7)
55–64	37 888	(11.3)	25 189	(14.5)	12 699	(7.8)
65–74	77 672	(23.2)	47 348	(27.3)	30 324	(18.7)
75–84	113 080	(33.7)	56 880	(32.8)	56 200	(34.7)
85+	86 263	(25.7)	31 078	(17.9)	55 185	(34.1)
Year of stroke						
2005	25 812	(7.7)	13 133	(7.6)	12 679	(7.8)
2006	24 822	(7.4)	12 429	(7.2)	12 393	(7.7)
2007	24 248	(7.2)	12 257	(7.1)	11 991	(7.4)
2008	24 541	(7.3)	12 514	(7.2)	12 027	(7.4)
2009	24 976	(7.5)	12 799	(7.4)	12 177	(7.5)
2010	25 569	(7.6)	12 891	(7.4)	12 678	(7.8)
2011	25 149	(7.5)	12 987	(7.5)	12 162	(7.5)
2012	24 815	(7.4)	12 792	(7.4)	12 023	(7.4)
2013	24 408	(7.3)	12 649	(7.3)	11 759	(7.3)
2014	23 613	(7.0)	12 340	(7.1)	11 273	(7.0)
2015	22 939	(6.8)	12 105	(7.0)	10 834	(6.7)
2016	21 863	(6.5)	11 594	(6.7)	10 269	(6.3)
2017	21 251	(6.3)	11 272	(6.5)	9979	(6.2)
2018	21 177	(6.3)	11 482	(6.6)	9695	(6.0)
Stroke type						
Hemorrhagic	40 969	(12.2)	22 131	(12.8)	18 838	(11.6)
Ischemic	286 634	(85.5)	147 588	(85.2)	139 046	(85.9)
Unspecified	7580	(2.3)	3525	(2.0)	4055	(2.5)
Consciousness						
Missing	4103	(1.2)	2155	(1.2)	1948	(1.2)
Alert	273 590	(81.6)	145 971	(84.3)	127 619	(78.8)
Drowsy	39 999	(11.9)	17 519	(10.1)	22 480	(13.9)
Unconscious	17 491	(5.2)	7599	(4.4)	9892	(6.1)
Institutional living						
Yes	31 945	(9.5)	11 736	(6.8)	20 209	(12.5)
No	301 322	(89.9)	160 496	(92.6)	140 826	(87.0)
Missing	1916	(0.6)	1012	(0.6)	904	(0.6)
Living alone						
Yes	165 024	(49.2)	63 273	(36.5)	101 751	(62.8)
No	167 393	(49.9)	108 583	(62.7)	58 810	(36.3)
Missing	2766	(0.8)	1388	(0.8)	1378	(0.9)
Dependent in p-ADL before stroke						
Yes	39 756	(11.9)	16 243	(9.4)	23 513	(14.5)
No	286 256	(85.4)	153 280	(88.5)	132 976	(82.1)
Missing	9171	(2.7)	3721	(2.1)	5450	(3.4)
Previous stroke						
Yes	81 928	(24.4)	44 002	(25.4)	37 926	(23.4)
No	250 580	(74.8)	127 914	(73.8)	122 666	(75.7)
Missing	2675	(0.8)	1328	(0.8)	1347	(0.8)
AF						
Yes	96 154	(28.7)	46 154	(26.6)	50 000	(30.9)
No	235 836	(70.4)	125 442	(72.4)	110 394	(68.2)
Missing	3193	(1.0)	1648	(1.0)	1545	(1.0)

(*Continued*)

Table 1. Continued

			Sex			
	Overall		Men		Women	
	N	(%)	n	(%)	n	(%)
Diabetes						
Yes	69 880	(20.8)	39 759	(22.9)	30 121	(18.6)
No	263 460	(78.6)	132 578	(76.5)	130 882	(80.8)
Missing	1843	(0.5)	907	(0.5)	936	(0.6)
Hypertensive medication at BL						
Yes	200 187	(59.7)	99 936	(57.7)	100 251	(61.9)
No	131 598	(39.3)	71 571	(41.3)	60 027	(37.1)
Missing	3398	(1.0)	1737	(1.0)	1661	(1.0)
Smoker						
Yes	42 565	(12.7)	24 880	(14.4)	17 685	(10.9)
No	255 395	(76.2)	129 299	(74.6)	126 096	(77.9)
Missing	37 223	(11.1)	19 065	(11.0)	18 158	(11.2)
Acute treatment						
Direct to SU						
Yes	248 479	(74.1)	130 867	(75.5)	117 612	(72.6)
No	85 340	(25.5)	41 583	(24.0)	43 757	(27.0)
Missing	1364	(0.4)	794	(0.5)	570	(0.4)
SU care						
Yes	294 705	(87.9)	154 267	(89.0)	140 438	(86.7)
No	35 681	(10.6)	16 637	(9.6)	19 044	(11.8)
Missing	4797	(1.4)	2340	(1.4)	2457	(1.5)
Reperfusion						
Yes	24 404	(8.5)	13 656	(9.3)	10 748	(7.7)
No	260 699	(91.0)	133 150	(90.3)	127 549	(91.8)
Missing	1400	(0.5)	711	(0.5)	689	(0.5)
Thrombolysis <30 min (2010–2018)						
Yes	4446	(23.4)	2529	(24.1)	1917	(22.6)
No	13 973	(73.7)	7633	(72.7)	6340	(74.9)
Missing	548	(2.9)	341	(3.2)	207	(2.4)
Secondary prevention						
BP lowering						
Yes	225 463	(76.6)	117 543	(76.0)	107 920	(77.3)
No	67 391	(22.9)	36 321	(23.5)	31 070	(22.3)
Missing	1348	(0.5)	750	(0.5)	598	(0.4)
Lipid lowering						
Yes	160 780	(62.0)	92 043	(67.8)	68 737	(55.7)
No	97 240	(37.5)	43 006	(31.7)	54 234	(43.9)
Missing	1186	(0.5)	644	(0.5)	542	(0.4)
Anticoagulants						
Yes	36 145	(50.4)	19 370	(55.6)	16 775	(45.5)
No	34 912	(48.7)	15 147	(43.5)	19 765	(53.6)
Missing	612	(0.9)	309	(0.9)	303	(0.8)
Antiplatelets						
Yes	167 144	(89.1)	90 333	(89.6)	76 811	(88.6)
No	19 518	(10.4)	10 028	(9.9)	9490	(10.9)
Missing	869	(0.5)	502	(0.5)	367	(0.4)
Outcome						
CF 7 d	22 324	(6.7)	10 202	(5.9)	12 122	(7.5)
CF 28 d	43 666	(13.0)	19 327	(11.2)	24 339	(15.0)
CF 90 d	60 290	(18.0)	26 696	(15.4)	33 594	(20.7)
ADL dependent	42 071	(20.2)	20 511	(18.1)	21 560	(22.8)

No. of patients (N) and frequency (%) in the whole study (overall) and in men and women 2005–2018. ADL indicates activities in daily living; AF, atrial fibrillation; BL, baseline; BP, blood pressure; CF, case fatality; and SU, stroke unit.

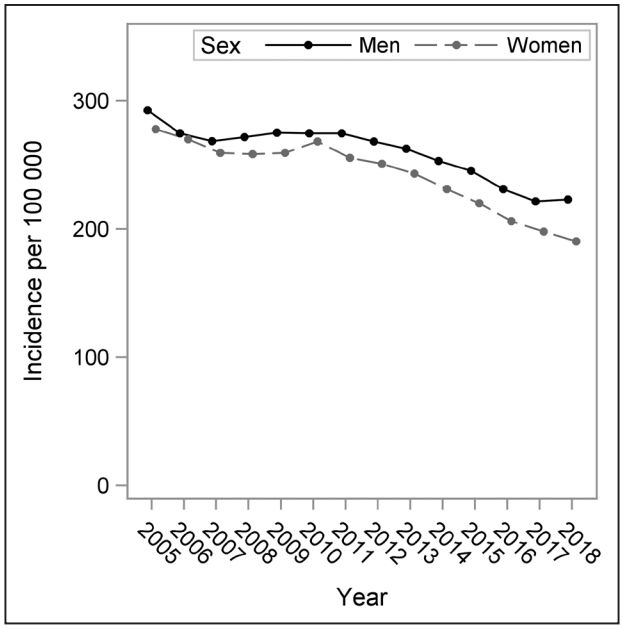

Figure 1. Estimated stroke incidence per 100 000 men and women, 2005–2018.

Long-term survival, up to 5 years after stroke, was presented by Kaplan-Meier survival curves, and the log-rank test was used to test if survival rate differed between men and women. CF and ADL dependency 3 months after stroke was also analyzed in an extended model, stratified by level of consciousness, including the additional independent variables stroke type, ADL dependency, living in institution before stroke, previous stroke, AF, diabetes, hypertensive medication, and smoking. Except for smoking status, the proportion of missing data was low (<3%), Table 1. A complete case approach was used, with the exception that missing information on smoking was included as a separate variable category.

We analyzed the data using SAS software version 9.4 for Windows (SAS Institute Inc, Cary, NC).

Ethical Considerations

The study was approved by the Swedish Ethical Review Authority (reference no. 2019-02777). In accordance with the Personal Data Act (Swedish law No. SFS 1998:204), no informed consent is needed to collect data from medical charts and inpatient records for quality registers.

RESULTS

Patient Characteristics

The study included 173 244 (51.7%) men and 161 939 (48.3%) women. The number of registered stroke events decreased continuously from 25 812 in 2005 to 21 177 in 2018. From 2005 to 2018, the incidence decreased in both men and women (from 293 to 223 per 100 000 men and from 278 to 191 per 100 000 women, Figure 1). As a result of the larger relative decrease in women (P<0.001), the proportion of men in the stroke population increased from 50.9% in 2005 to 54.2% in 2018.

During the study period, men were on average 4.9 years younger than women at stroke onset (73.5 versus 78.3 years in 2005 and 73.0 versus 77.5 years in 2018). Accordingly, men were also less often living alone, living in an institution, or being dependent in ADL at stroke onset than were women (Table 1). In addition, previous stroke, diabetes, and smoking were more common in men, while AF and antihypertensives were more common in women. A lower proportion of men than women had lowered consciousness at hospital arrival (Table 1).

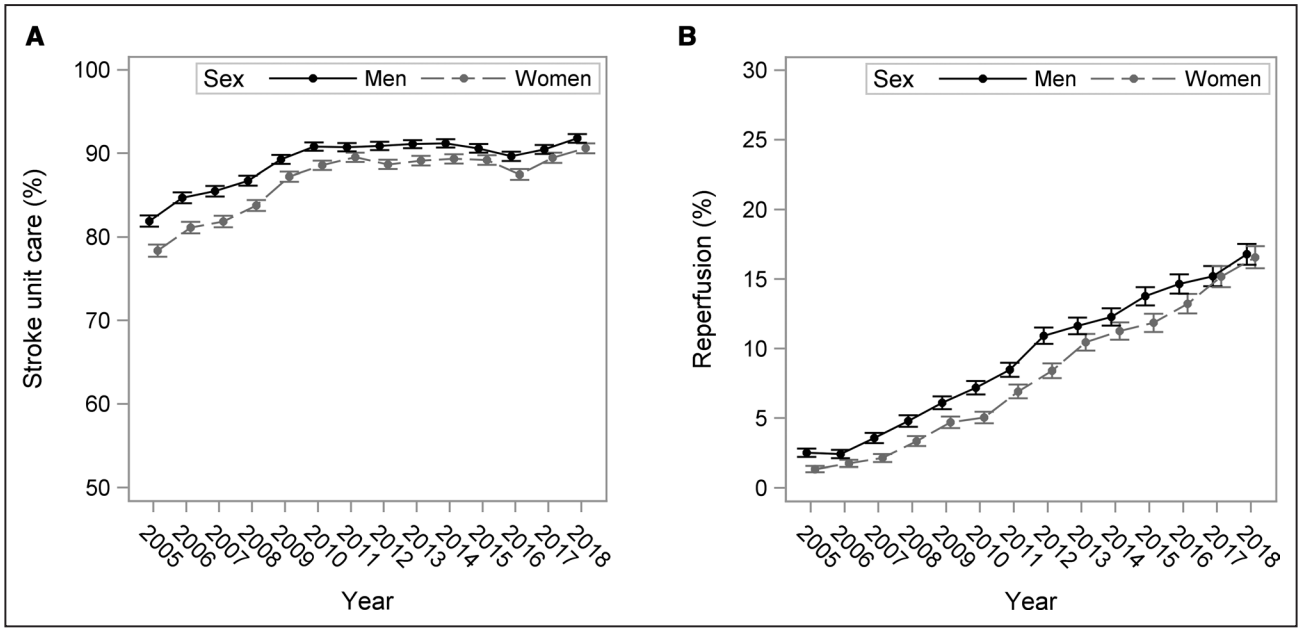

Figure 2. Acute stroke care 2005–2018.
A, Stroke unit care (%) and (**B**) reperfusion therapy (%) in patients with ischemic stroke, with 95% CIs.

Acute Care

From 2005 to 2018, the proportion of patients treated in a stroke unit increased in both men and women at a similar rate (Figure 2 and Table 2). In 2018, the probabilities of stroke unit care were similar in men and women (age-adjusted proportion 92.6% versus 92.4%). However, men were somewhat more likely to be admitted directly to a stroke unit, a difference that persisted in 2018 (age-adjusted proportion 79.6% versus 78.2%, Table 2).

In 2005, a higher proportion of men received reperfusion therapy (age-adjusted proportion 2.3% versus 1.4%, Table 2). This proportion increased in both sexes, but more rapidly in women (Figure 2) and in 2018, this difference was reversed (15.1% versus 16.9%). Of the patients who received thrombolysis, the proportion receiving treatment within 30 minutes also increased at a similar rate in men and women (Table 2). Men had a higher probability of thrombolysis within 30 minutes than women (age-adjusted proportion 2010–2018: 20.5% versus 18.4%). This difference was not explained by women's higher probability of living alone (data not shown).

Secondary Prevention

A similar proportion of men and women was prescribed antihypertensives at discharge. The age-adjusted proportion increased during 2005 to 2018 (from 73.9% to 80.3% in men and from 74.7% to 79.8% in women, Table 2).

In patients with ischemic stroke, men were more likely to be prescribed statins at discharge compared with women (age-adjusted proportions 2005–2018: 67.5% versus 62.4%). In this age-adjusted analysis, a more pronounced difference was found in patients with AF (53.0% versus 45.6%) than in patients without (72.8% versus 68.6). Even if the probability of treatment increased more

Table 2. Estimated Proportions (%) by a Generalized Linear Model Including the Independent Variables Sex, Year, Sex-by-Year Interaction and Adjusting for Age, and Age^2

Variable	Sex	Overall proportion 2005–2018 (95% CI)	*P* overall sex difference	Proportion 2005 (95% CI)	Proportion 2018 (95% CI)	*P* sex difference varies over time
Acute care						
Direct to SU	Men	75.5 (75.3–75.7)	<0.001	66.0 (65.2–66.8)	79.6 (78.8–80.3)	<0.306
	Women	74.3 (74.0–74.5)		64.3 (63.4–65.1)	78.2 (77.4–79.0)	
SU care	Men	90.2 (90.1–90.4)	<0.001	84.4 (83.7–85.0)	92.6 (92.1–93.1)	0.224
	Women	89.6 (89.5–89.8)		83.1 (82.5–83.8)	92.4 (91.8–92.8)	
Reperfusion	Men	7.1 (7.0–7.3)	<0.001	2.3 (2.0–2.6)	15.1 (14.5–15.8)	<0.001
	Women	6.5 (6.4–6.7)		1.4 (1.2–1.7)	16.9 (16.1–17.8)	
Thrombolysis <30 min (2010–2018)	Men	20.5 (19.5–21.5)	0.005	5.6 (4.1–7.6)	36.6 (34.0–39.2)	
	Women	18.4 (17.3–19.5)		4.7 (3.1–7.0)	34.2 (31.5–37.0)	0.674
Secondary prevention						
BP lowering	Men	77.5 (77.3–77.8)	0.357	73.9 (73.0–74.7)	80.3 (79.5–81.1)	0.013
	Women	77.7 (77.5–77.9)		74.7 (73.9–75.5)	79.8 (78.9–80.6)	
Lipid lowering	Men	67.5 (67.2–67.8)	<0.001	36.9 (35.9–37.8)	83.7 (82.8–84.5)	0.027
	Women	62.4 (62.1–62.7)		32.3 (31.3–33.3)	81.2 (80.3–82.1)	
Anticoagulants	Men	53.4 (52.8–54.0)	<0.001	31.2 (29.4–33.1)	78.6 (76.8–80.2)	<0.001
	Women	51.3 (50.7–51.9)		26.7 (25.0–28.4)	81.9 (80.2–83.4)	
Antiplatelets	Men	90.0 (89.8–90.2)	<0.001	88.6 (87.8–89.3)	90.4 (89.6–91.1)	0.738
	Women	89.4 (89.2–89.6)		88.2 (87.4–89.0)	90.0 (89.2–90.8)	
Outcome						
CF 7 d	Men	5.9 (5.8–6.1)	0.002	5.8 (5.4–6.2)	5.6 (5.2–6.1)	0.062
	Women	6.2 (6.1–6.3)		6.1 (5.7–6.5)	6.2 (5.8–6.7)	
CF 28 d	Men	11.1 (10.9–11.2)	<0.001	11.3 (10.7–11.8)	10.9 (10.3–11.5)	0.301
	Women	11.6 (11.4–11.8)		11.1 (10.6–11.6)	11.4 (10.9–11.9)	
CF 90 d	Men	15.3 (15.2–15.5)	<0.001	16.3 (15.7–17.0)	14.7 (14.0–15.4	0.604
	Women	15.9 (15.7–16.1)		16.2 (15.6–16.8)	15.0 (14.4–15.7)	
ADL dependent	Men	18.0 (17.8–18.3)	<0.001	20.6 (19.7–21.5)	16.0 (15.1–16.9)	0.588
	Women	19.1 (18.8–19.4)		21.2 (20.3–22.1)	16.4 (15.4–17.3)	

Estimated proportions during the whole period 2005–2018, test if overall proportion differs between men and women, estimated proportion in 2005 and 2018, respectively, test if the trend differs between men and women 2005–2018 (all years were included in the model but proportions only shown for 2005 and 2018). For example, reperfusion was more common in men than in women, overall 2005–2018 (*P* overall sex diff <0.001) and the differences changed over the time analyzed (*P* sex-trend varies over time <0.001) to be more common in women compared with men in 2018. ADL indicates activities in daily living; CF, case fatality; and SU, stroke unit.

rapidly in women than in men (Figure 3), men still had a higher probability of treatment in 2018 (age-adjusted proportion 83.7% versus 81.2%, Table 2).

In patients with ischemic stroke and AF, men had a higher likelihood of anticoagulant treatment than women in the beginning of the study period (age-adjusted proportion 53.4% versus 51.3% in 2005). This rate increased more rapidly in women (Figure 3) and by 2018 the age-adjusted proportion was higher in women (78.6% in men versus 81.9% in women, Table 2). The proportion receiving antiplatelets remained at a similar level, with no major differences between men and women during the study period.

Outcome

ADL-function and survival after stroke improved from 2005 to 2018 in both men and women (Table 2, Figure 4A and 4B). Men had better survival than women up to 5 years after stroke ($P<0.001$). The difference appeared already in the acute phase, increased up to 3 months after stroke where after the difference remained stable (Figure 4B). The main difference between men and women was explained by women's higher age (Figure 4C). After age adjustment, only a small difference between men and women remained (CF 7 days: 5.9% versus 6.2%, CF 28 days: 11.1% versus 11.6%, and CF 90 days: 15.3 versus 15.9). After additional adjustments for ADL dependence, living in institution, and cardiovascular risk factors at stroke onset, survival was similar in men and women (CF 7 days: 5.9% versus 6.2%, CF 28 days: 11.1% versus 11.6%, and CF 90 days: 15.3% versus 15.9%). In men and women with the same level of consciousness on arrival to hospital, men had lower survival than women (eg, CF 7 days: 1.5% versus 1.3%, CF 28 days: 4.1% versus 3.7%, and CF 90 days: 7.2% versus 6.6%, in patients who were fully conscious).

In survivors, ADL 3 months after stroke improved at a similar rate in men and women during the study period (Figure 4A). The observed difference in ADL dependency between men and women 3 months after stroke was to a large extent explained by the age difference (Table 2). After age adjustment, the estimated difference between men and women was small (20.6% versus 21.2% in 2005 and 16.0% versus 16.4% in 2018). However, this difference remained statistically significant after additional adjustments ($P<0.001$).

DISCUSSION

In this large, nation-wide, observational study covering 14 years, we observed a decreasing stroke incidence. There have been major improvements in acute care, secondary preventive care, and outcome 3 months after stroke. Women were on average 5 years older at stroke onset, and after age adjustments, our study demonstrates diminished sex dissimilarities. Yet, women were less often prescribed statins and were less often admitted directly to a stroke unit.

Despite an aging population and an increase in total population of 1.1 million inhabitants (or 12%) between 2005 and 2017,[2,3] we report a decrease in total numbers of stroke in Sweden from 25 812 to 21 177 between 2005 and 2018, which is to be expected from a high-income country.[23–26] In low- and middle-income countries, on the contrary, the incidence is increasing.[7,27] Why stroke incidence is declining is debated, better treatment of risk factors, modified lifestyle factors such as less smoking, and an overall better health

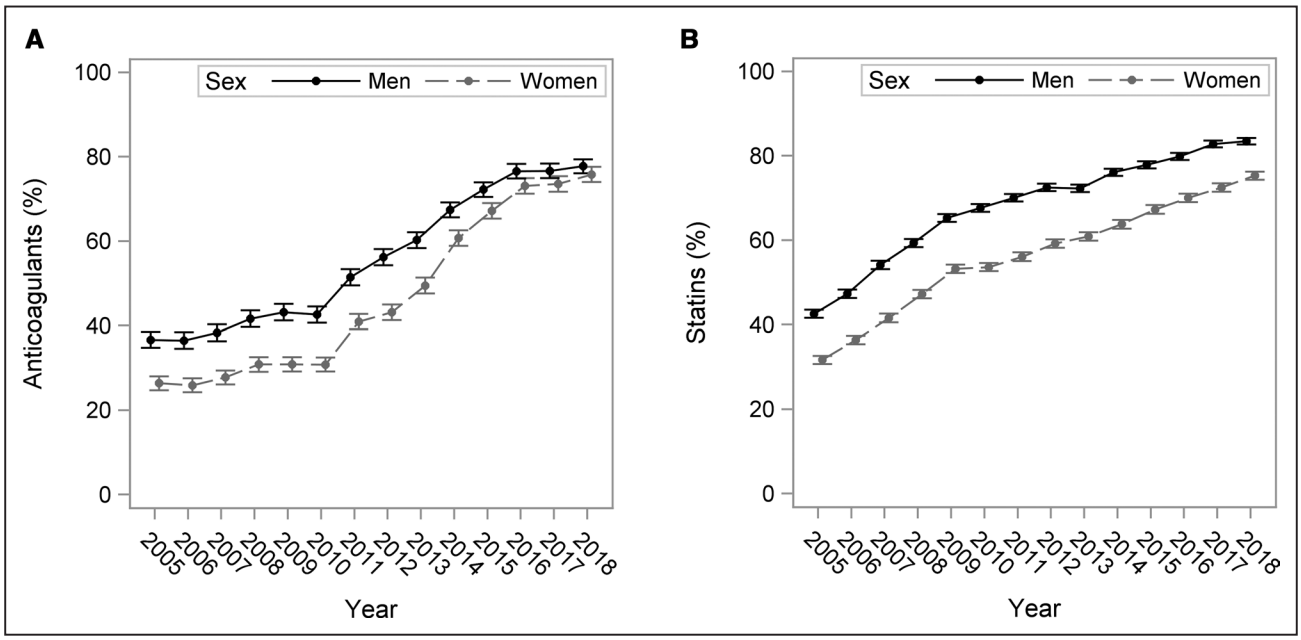

Figure 3. Secondary prevention 2005–2018.
A, Anticoagulant treatment at discharge (%) in patients with ischemic stroke and atrial fibrillation and (**B**) lipid-lowering drugs at discharge (%) in patients with ischemic stroke, with 95% CIs.

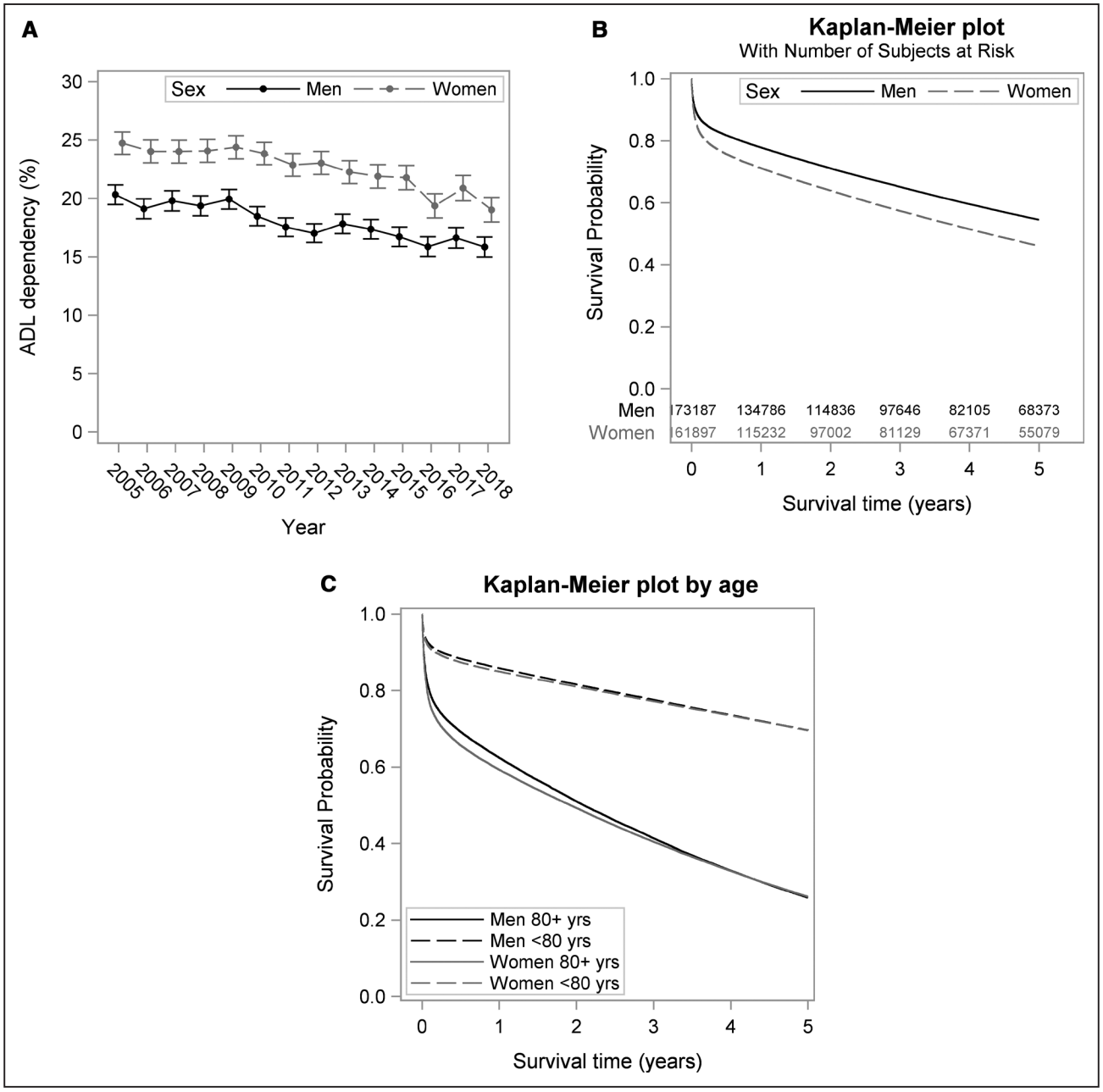

Figure 4. Stroke outcome 2005–2018.
A, Activities in daily living (ADL) dependency (%) 3 mo after stroke in patients who were independent before stroke, with 95% CIs, (**B**) Kaplan-Meier survival curves up to 5 y after stroke with number of patients at risk for each year, and (**C**) Kaplan-Meier survival curves up to 5 y after stroke, stratified by age above/below 80 y.

has been suggested.[5,28] We also observed a more pronounced decrease in the incidence of stroke among women, which is contrary to what is described in other studies.[29]

Acute Care

Stroke unit care is demonstrated to decrease CF and increase the chance of a good outcome.[30,31] Access to stroke unit care was equally distributed between women and men during the entire study period (>92%), yet, women were somewhat less likely than men to be admitted directly to the stroke unit, hence, missing the opportunity of evidence based care at stroke onset.

In 2005, the proportion of patients with acute ischemic stroke treated with reperfusion therapy was low, and more so in women in comparison with men. During the study period, the use of reperfusion therapy increased more rapidly in women; in 2018, the age-adjusted proportion of treated patients was even higher in women than in men. In conjunction with the publishing of the IST-3 study (The Third International Stroke Trial) on reperfusion therapy, including 53% patients over the age

of 80 years,[32] a pronounced increase of thrombolysis in patients in that age group was seen in our study population. The opening of reperfusion therapy to patients aged above 80 years increased the opportunity for women to get access to the treatment as 51% of the women compared with 31% of men with ischemic stroke in our study were above 80 years.

Secondary Prevention

The proportions of women prescribed anticoagulants and statins at discharge were lower compared with the proportions of men. Despite the beneficial preventive effect of anticoagulants for patients with ischemic stroke and AF, several studies point out an undertreatment in both sexes, but particular among women.[33–35] We report a considerable increase in anticoagulant utilization, which was more pronounced in women. This is in line with data shown in other studies.[35] The increased use, and diminished sex gap, has coincided with the introduction of nonvitamin K antagonist oral anticoagulants, and with that, a large focus on treatment of AF. A US study found no sex difference in the probability of the patient filling nonvitamin K antagonist oral anticoagulants versus warfarin and found women more likely to start nonvitamin K antagonist oral anticoagulants in the later part of the time period studied, that is, 2015 to 2017 as compared with 2010 to 2012.[36] In our study, women were less often discharged with statins; despite a steeper increase of statin use in women than in men, the difference was consistent throughout the study period. Similar findings have been presented in earlier Swedish and international studies.[37–40] An interaction between AF and statin prescriptions, suggests that women getting anticoagulation, are less likely to be prescribed statins. Still, in Sweden, statins are recommended after ischemic stroke, regardless of stroke cause including AF.[22]

Outcome

Considerable variation in CF differences between women and men is previously reported.[41] As in most studies, and during our entire study period, women in our study were older. In addition, they had a higher stroke severity, higher proportion of cardioembolic strokes were more often living in an institution, and were dependent in ADL at stroke onset. The difference in survival between men and women was small after adjustment for women's higher age at stroke onset and even reversed if we considered that women had more severe strokes.

Strength and Limitations

An advantage of using registry-based data is the possibility to include a large, unselected cohort. The Riksstroke started in 1994 and is a national quality register with excellent coverage and data on >90% of all stroke events in Sweden are registered.[1] The data collected in Riksstroke has previously been validated.[42] Since the National Institutes of Health Stroke Scale was missing for almost 50% of the cases in Riksstroke,[1] we used level of consciousness as a proxy for stroke severity. However, using level of consciousness, as a proxy for severity, has been shown to be a good predictor of CF.[43] As in all observational studies, establishing causation between measures and outcomes is not possible. Furthermore, there may be unmeasured confounding which we have not corrected for.

Conclusions

Incidence of stroke has decreased, more so in women than in men, resulting in a larger proportion of the stroke group being male. Acute care, secondary prevention, and outcome have improved in both men and women. The sex gap in care and secondary prevention has diminished, albeit it has not yet fully disappeared. When taking the stroke severity and age at stroke onset into account functional outcome and survival first 3 months in recent observation period were similar in men and women. In all, we see encouraging changes, although there are still room for improvement to reduce number of stroke and decrease the impact of the disease.

ARTICLE INFORMATION

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Supplemental Materials

Online Figure I
Online Table I

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